

ML Specification Reports — 15 Playground-Series Competitions

Each competition is documented as a **report** against the 顏佐榕 ML-spec framework, satisfying the summer-project plan's Goal 1: **(甲) 競賽目的** — **what + why** (Overview + "Why it matters") and **(乙) 五大流程元件** — Data / Models / Training / Inference / Evaluation. Presentation is narrative-first with tables only where genuinely tabular; every figure is grounded in that competition's `facts.json` / `config.yaml` / `eda_summary.json` / `STATUS.md` / `experiments_tree_v3.json`, and DL-only items with no tabular-GBDT analogue are marked *N/A (reason)*.

Each report has a Markdown source (`playground-series-<comp>.md`) and an individual academic-serif PDF (`pdf/playground-series-<comp>.pdf`). Rubric validation: `RUBRIC.md` (all 15 pass 8/8 plan-aligned checks).

#	Competition	Task	Metric	Our best (CV)	NVIDIA	Winner
1	s3e1	California Housing — median house value	RMSE ↓	0.5571	0.5086	NVIDIA
2	s3e3	Employee attrition	AUC ↑	0.8378	0.8758	NVIDIA
3	s3e5	Wine quality (ordinal)	QWK ↑	0.5677	0.5478	our agent
4	s3e7	Hotel reservation cancellation	AUC ↑	0.8999	0.9014	tie
5	s3e9	Concrete compressive strength	RMSE ↓	12.07	12.026	tie
6	s3e11	Media campaign cost	RMSLE ↓	0.29565	0.2956	tie
7	s3e14	Wild blueberry yield	MAE ↓	340.6	337.28	NVIDIA
8	s3e16	Crab age	MAE ↓	1.3356	1.3387	our agent
9	s3e19	Forecast mini-course sales	SMAPE ↓	10.02	13.72	our agent
10	s3e20	Rwanda CO2 emission	RMSE ↓	21.15	19.648	NVIDIA
11	s4e1	Bank customer churn	AUC ↑	0.894393	0.8984	NVIDIA
12	s4e11	Depression prediction	Accuracy ↑	0.9402	0.9399	tie
13	s5e10	Road accident risk	RMSE ↓	0.05597	0.05609	tie
14	s6e1	Student exam-score	R ² ↑	0.78718	0.7863	tie
15	s6e2	Heart disease	AUC ↑	0.95553	0.9554	tie

Record vs NVIDIA reproduce-agent: **3 our-agent wins / 7 ties / 5 NVIDIA wins**. Scores are local CV (all competitions closed → no live leaderboard); benchmark methodology and caveats (de-leaking, unified CV schemes) are documented per report and in the four-axis evaluation.